The integers are $\mathbb{Z} = \{a : a = [m,n],\ m,n \in \mathbb{N}\}$

$[m,n] = \{(a,b) \in \mathbb{N} \times \mathbb{N} : (a,b) \sim (m,n)\}$

This is a set

$\Delta(a,b) = \Delta(m,n)$

But also an element of integers. $(5,7) \neq (1,3)$ but $[5,7] = [1,3] = -2$

If $a = [m,n]$ ~~and~~ we define $-a$ to be $[n,m]$

$a + -a = [m,n] + [n,m] = [m+n, n+m] = [m+n, m+n] = [0,0]$

Thm 7

$\Delta(m+n, m+n) = (0,0) = \Delta(0,0)$

$[5,7] + [0,0] = [5+0, 7+0] = [5,7]$.

Define Subtraction If $a = [m,n]$ and $b = [r,s]$

then $a - b \overset{def}{=} a + (-b) = [m,n] + [s,r] = [m+s, n+r]$ by def

Ex. $a = [5,7]$; $b = [1,2]$ then $a - b = [5,7] + [2,1] = [5+2, 7+1]$

$= [7,8]$

$\Delta(0,1) = (0,1)$ $\Delta(7,8) = (0,1)$

Def 11. The function $\mathbb{N} \to \mathbb{Z}$ is given by $n \mapsto [n,0]$ is called canonical embedding. We will denote $[n,0]$ as n. $\mathbb{N}$ lives inside $\mathbb{Z}$

Thm 19. embedding is injective,

Thm 20; Addition in $\mathbb{N}$ agrees with addition in $\mathbb{Z}$. "If $n = [n,0]$ and $m = [m,0]$ $\to [n+m, 0]$

Proof; We compute $n+m$ maps to $[n+m, 0]$ under embedding.

Likewise, $n+m$ in $\mathbb{Z} = [n,0] + [m,0] = [n+m, 0]$

$\Delta(a,b) = \Delta(m,n)$

Thm 21: Subtraction in $\mathbb{N}$ agrees with subtraction in $\mathbb{Z}$

Thm 22: If $a \in \mathbb{Z}$, exactly one of the following occurs: $a = \emptyset = [0,0]$, a is 0, a is $+$, a is $-$

or $a = n \in \mathbb{N}^+$, or $-a = n \in \mathbb{N}^+$.

Proof. Let $a = [m,k] \in \mathbb{Z}$

By thm 7, $[m,k] = [\Delta(m,k)]$. By def of trichotomy in $\mathbb{N}$,

exactly one occurs: $[m,k] = (0,0)$, $[m,k] = (m-k, 0)$, $[m,k] = (0, k-m)$

Thus $[m,k] = [0,0]$, $[m,k] = [m-k, 0]$ or $[m,k] = [0, k-m]$

$a = 0$ $\quad$ $a = m-k \in \mathbb{N}^+$ $\quad$ $-a = k-m \in \mathbb{N}^+$